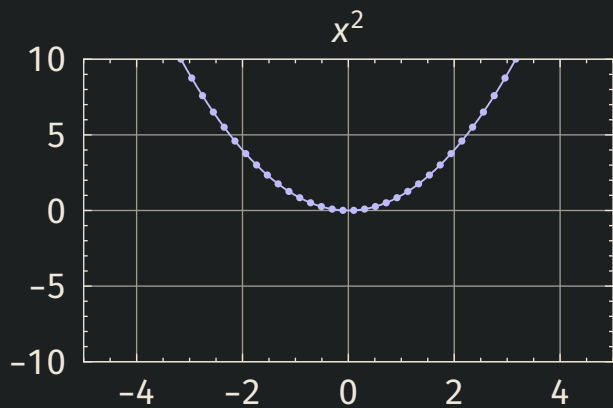


Functions

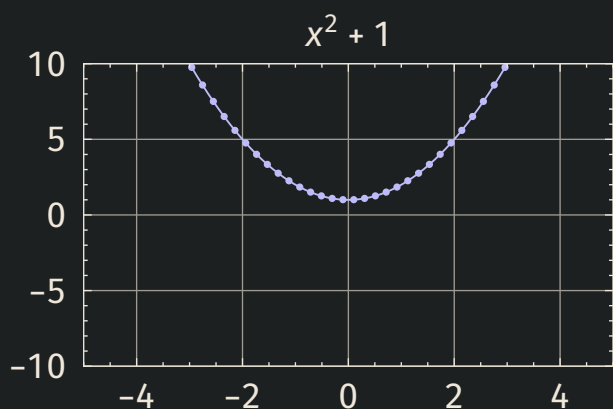
1. Transformation of functions

1.1. Translations

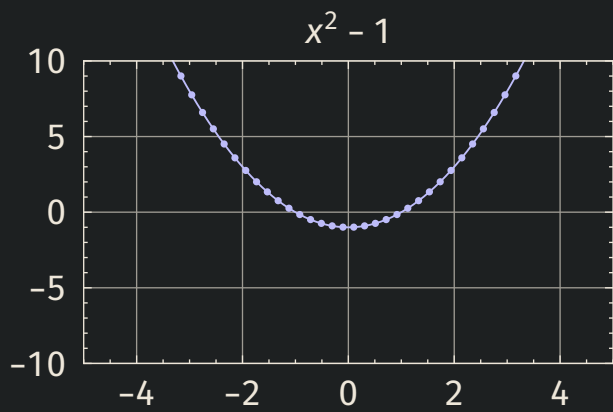
- $f(x) = x^2 \rightarrow$ Parent Function



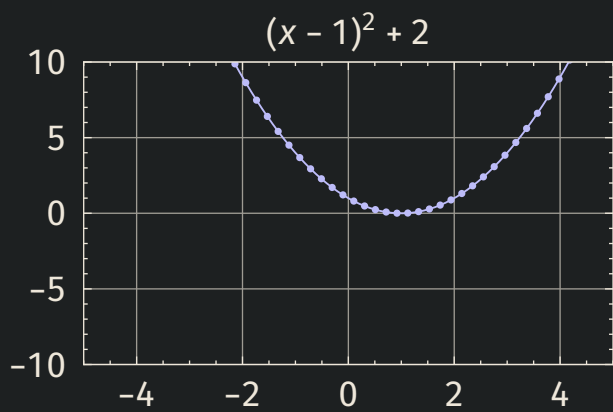
- $f(x) + c \rightarrow$ Shifts the graph c units up:



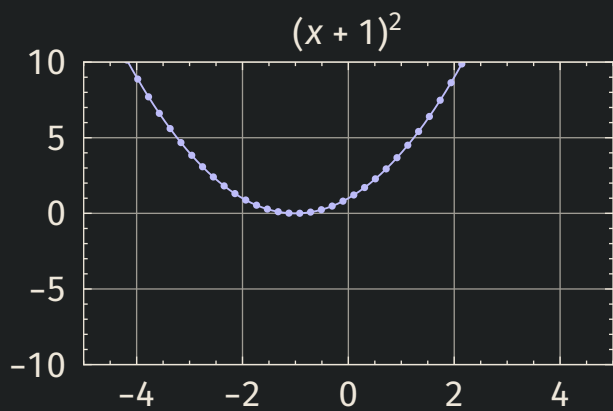
- $f(x) - c \rightarrow$ Shifts the graph c units down



- $f(x - c) \rightarrow$ Shifts the graph c units right

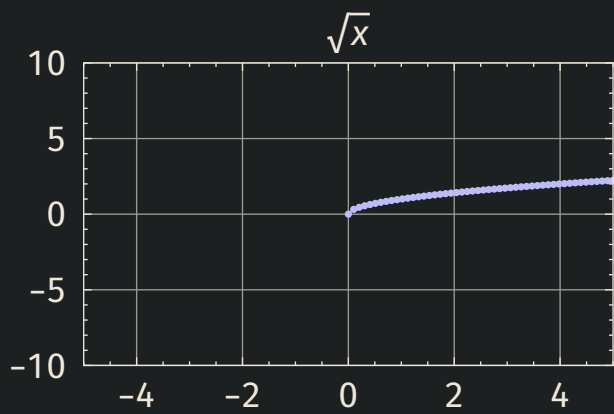


- $f(x + c) \rightarrow$ Shifts the graph c units left

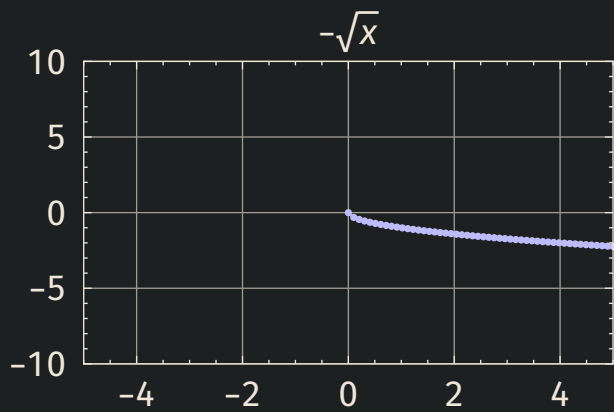


1.2. Reflections

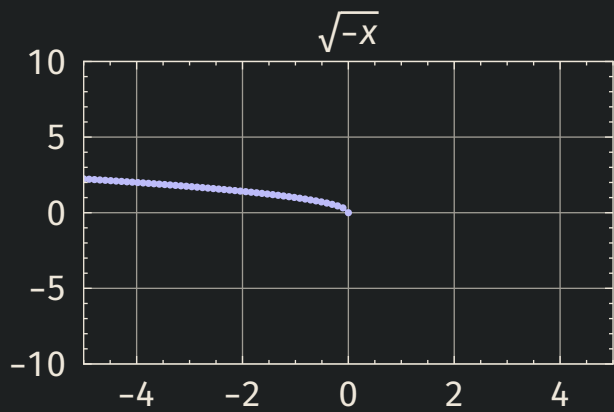
- $f(x) = \sqrt{x} \rightarrow$ Parent Function



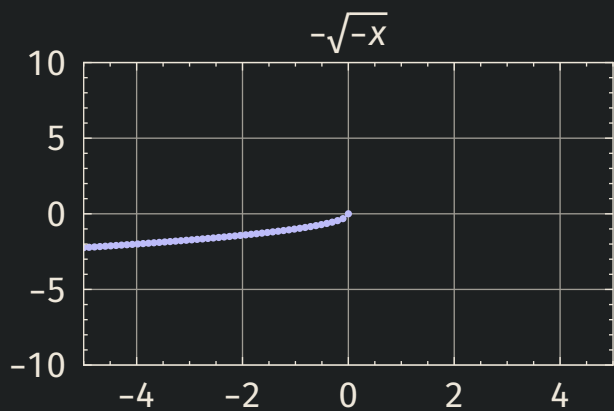
- $-f(x) \rightarrow$ Flips graph over the x-axis



- $f(-x) \rightarrow$ Flips graph over the y-axis

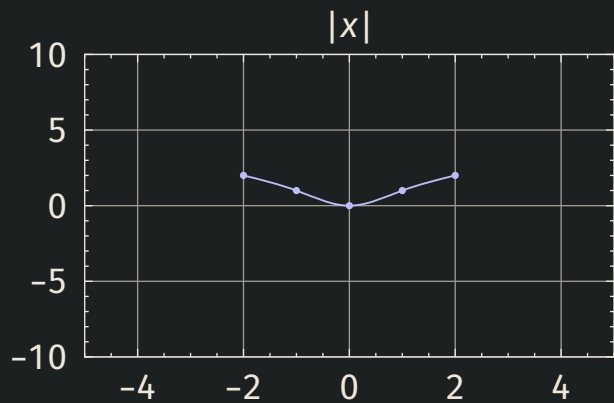


- $-f(-x) \rightarrow$ 180° rotation over the origin

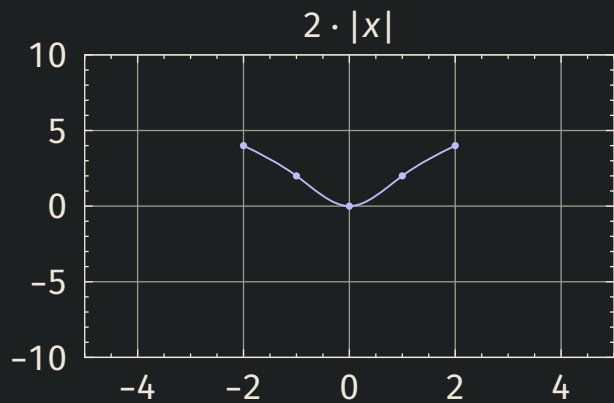


1.3. Dilations

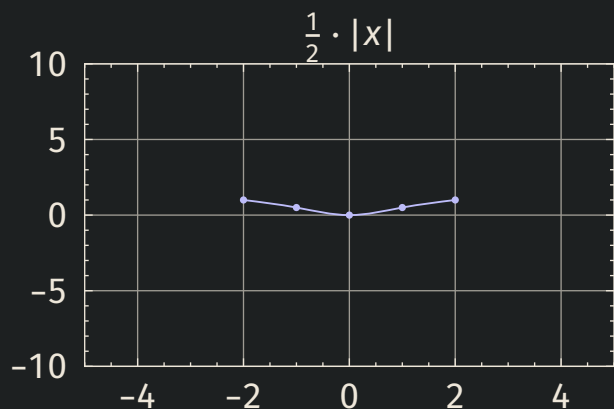
- $f(x) = |x| \rightarrow$ Parent Function



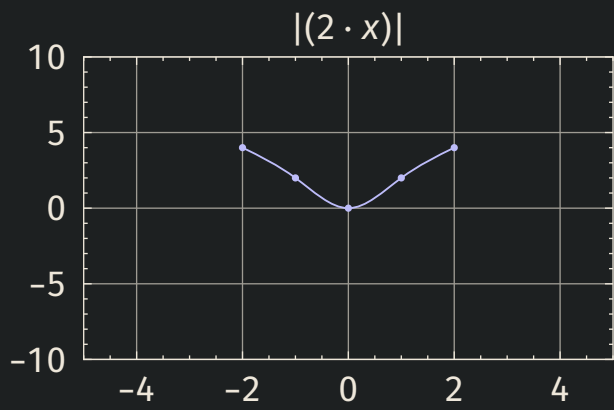
- $a \cdot f(x) \rightarrow a > 1$, Graph is stretched vertically



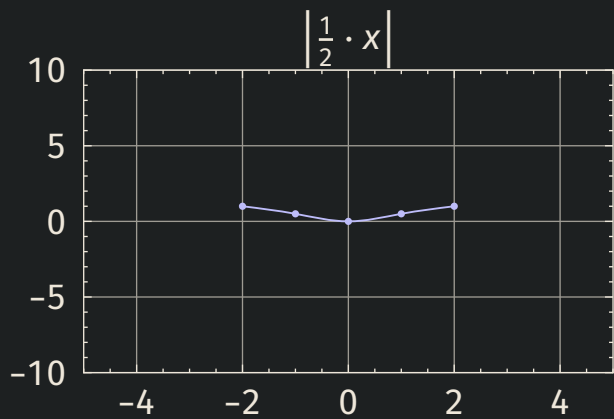
- $a \cdot f(x) \rightarrow 0 < a < 1$, Graph is shrunk vertically



- $f(a \cdot x) \rightarrow a > 1$, Graph is shrunk horizontally

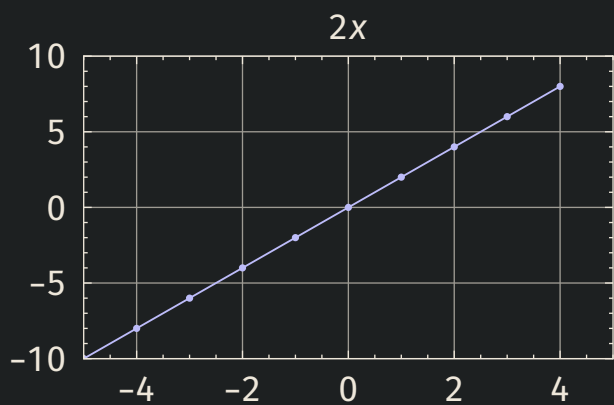


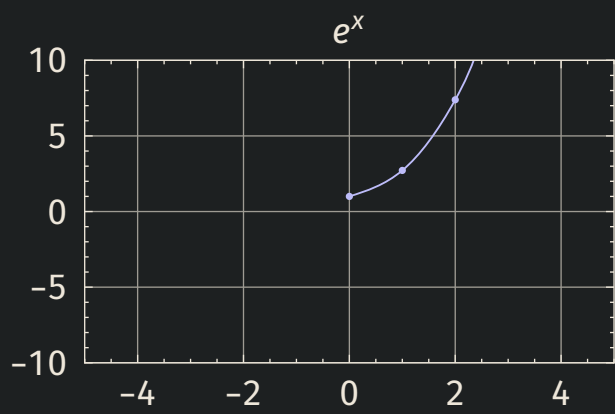
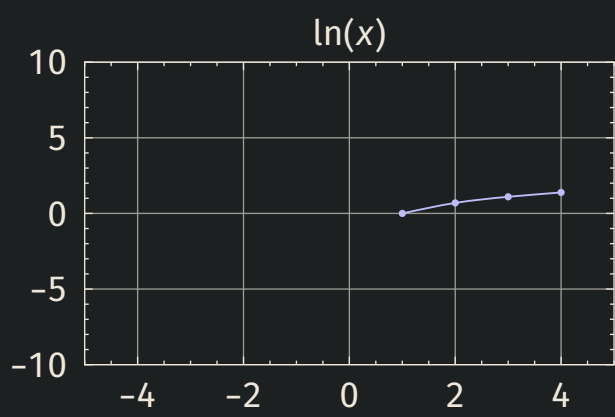
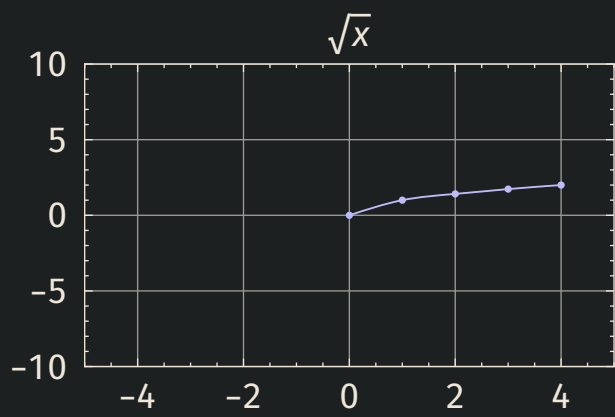
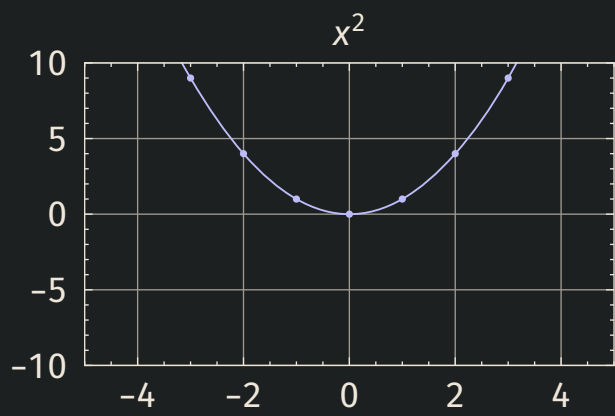
- $f(a \cdot x) \rightarrow 0 < a < 1$, Graph is stretched horizontally



2. Graphs

2.1. Common Graphs





2.2. Trig Functions

